3/25/2022

Friday, March 25, 2022 10:45 AM

M168_Notes_7.5

7.5 Conditional Probability and Independent Events

A survey is done of people making purchases at a gas station. Most people buy gas (Event A) or a drink (Event B).

	buy drink (B)	no drink (B^c)	total
buy gas (A)	45	25	70
no gas (A^c)	20	10	30
total	65	35	100

What is the probability that a person bought gas and a drink?

$\frac{45}{100}$

What the probability that *a person who buys a drink* also buys gas? In other words, **given** that a person *bought a drink* (B), what is the probability that they bought gas (A)?

$\frac{45}{65}$

NOTATION: P(A|B) = the probability of A given B

The ***conditional probability*** of event E given event F is

$$P(E|F) = \frac{n(E \cap F)}{n(F)} = \frac{n(E \cap F)/n(S)}{n(F)/n(S)} = \boxed{\frac{P(E \cap F)}{P(F)}}$$

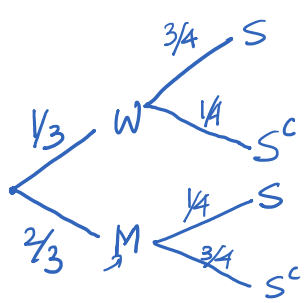
What is the probability that *a person who buys gas* also buys a drink?

$\frac{45}{70}$

PRODUCT RULE: $P(E \cap F) = P(E)P(F|E)$

TREE DIAGRAMS

At a party, 1/3 of the guests are women. 75% of the women wore sandals and 25% of the men wore sandals. What is the probability that a person chosen at random at the party is a man wearing sandals, $P(M \cap S)$? What is the probability that a randomly chosen guest is wearing sandals?



$$P(M \cap S) = \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{6}$$

$$P(W \cap S) + P(M \cap S) = P(S)$$

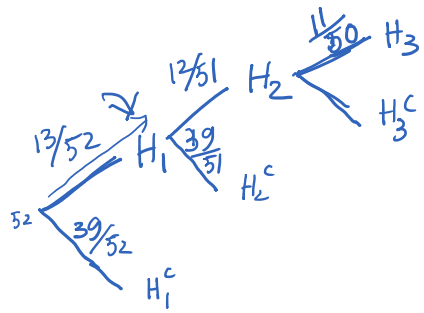
$$P(W \cap S) = \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

$$P(S) = \frac{1}{4} + 1/6 = \frac{5}{12}$$

$$P(M \cap S^c) = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}.$$

A finite stochastic process is one in which the next stage of the process depends on the state you are in for the previous stage.

Consider drawing 3 cards from a standard deck of 52 cards without replacement.
(a) What is the probability that the three cards are hearts?
(b) What is the probability that the third card drawn is a heart given the first two cards are hearts?



$$P(H_1) = \frac{13}{52}$$

$$P(H_2 \mid H_1) = \frac{12}{51}$$

$$P(H_3 \mid H_2) = \frac{11}{50}$$

$$P(H_1 \cap H_2 \cap H_3) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50}$$

– What is the prob. the 2nd card is a diamond given the first card is a heart.

$$P(D_2 \mid H_1) = \frac{13}{51}$$

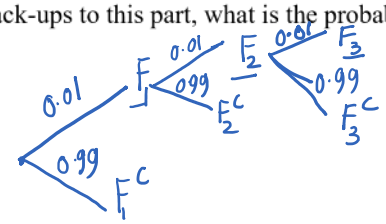
INDEPENDENT EVENTS: $\boxed{P(E|F) = P(E)}$

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)} = P(E) \Rightarrow$$

$\boxed{P(E \cap F) = P(E) \cdot P(F)}$ *iff* E and F are independent

Example

A certain part on airplane has a 1% chance of failure. If they carry two back-ups to this part, what is the probability that they will all fail?



They fail independent of each other.

$$P(F_1 \cap F_2 \cap F_3) \overset{\downarrow}{=} P(F_1) \cdot P(F_2) \cdot P(F_3)$$

$$= 0.01 \times 0.01 \times 0.01 = \frac{1}{10^6}$$

Example

A medical experiment showed the probability that a new medicine was effective was 0.75, the probability of a certain side effect was 0.4 and the probability for both occurring is 0.3. Are these events independent?

$$P(S) = 0.4$$

$$P(E \cap S) = 0.3$$

$$P(E) = 0.75$$

$$P(E \cap S) \overset{?}{=} P(E) \cdot P(S)$$

$$0.3 = 0.75 \times 0.4$$

$\Rightarrow$ Yes they are independent.